



Continuous Assessment One

MACHINE LEARNING AND PATTERN RECOGNITION

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Assessment:

Item	Description
Module Title:	Machine Learning and Pattern Recognition
Module Code:	B9AI102
Module Lecturer (s):	Assem Abdel-Hak
Programme/Cohort:	MSc Artificial Intelligence
Method of Assessment:	Continuous Assessment
Percentage (%) Weighting:	40%
MIMLOs being assessed:	1,2,3
Assessment Number:	1
Individual/Group:	Individual
Issue Date:	Week 4
Submission Date:	Week 7
Feedback Date:	Week 9
Feedback Strategy:	Recorded verbal

Assignment Task

Dataset

Each row in Bank.csv corresponds to a bank's credit card customer.

Relevant information about this dataset is given below:

Number of Instances: 4522

Number of Attributes: 16 independent variables + 1 target variable

Independent Variables: Input variables (bank client data):

1. Age (numeric)
2. job : type of job (categorical:
"admin.", "unknown", "unemployed", "management", "housemaid", "entrepreneur", "student",
"blue-collar", "self-employed", "retired", "technician", "services")
3. Marital : marital status (categorical: "married", "divorced", "single")
4. Education (categorical: "unknown", "secondary", "primary", "tertiary")
5. default: has credit in default? (binary: "yes", "no")
6. balance: average yearly balance, in euros (numeric)
7. housing: has housing loan? (binary: "yes", "no")
8. Loan: has personal loan? (binary: "yes", "no")
9. Contact: contact communication type (categorical: "unknown", "telephone", "cellular")
10. Day: last contact day of the month (numeric)
11. Month: last contact month of year (categorical: "jan", "feb", "mar", ..., "nov", "dec")
12. Duration: last contact duration, in seconds (numeric)
13. Campaign: number of contacts performed during this campaign and for this client (numeric, includes last contact)
14. Pdays: number of days that passed by after the client was last contacted from a previous campaign (numeric, -1 means client was not previously contacted)
15. Previous: number of contacts performed before this campaign and for this client (numeric)

16. Poutcome: outcome of the previous marketing campaign (categorical: "unknown","other","failure","success")

Output variable (desired target):

1. Y - has the client subscribed a term deposit? (binary: "yes","no").

Task

The bank wants to use a classification model that can predict whether the client has subscribed a term deposit? Construct a suitable classification model for the bank by implementing both random forest and support vector classification algorithms in Python.

Deliverables

1. A PDF Report (named as Report_Firstname_Surname.pdf) **(60% weighting)**
 2. A Python Code File (named as Code_Firstname_Surname.py) **(40% weighting)**
- Submit both in a zipped folder (named Firstname_Surname.zip)

There is no prescribed word-count for the report. It will be assessed on quality, and not quantity of content.

Your code and analysis should cover the following:

1. Data Preparation

(What steps would you take to prepare your data? Discuss your approach.)

Key Assessment Areas:

- Data preprocessing pipeline
- Handling missing values, encoding, and scaling
- Justification for chosen preprocessing steps

2. Machine Learning Implementation & Model Hyperparameter Tuning

(Which hyperparameters would you tune and why? How would you tune them?)

Key Assessment Areas:

- Implementation of multiple classification algorithms
- Hyperparameter tuning strategy (e.g., grid search, random search)
- Comparison of model performance

3. Choice of Evaluation Metric

(Which metric would be suitable for model evaluation and why?)

Key Assessment Areas:

- Justification for chosen evaluation metric(s)
- Alignment between the business/context goal and the metric
- Evidence of implementation and correct usage

4. Overfitting Avoidance Mechanism

(What would you use to avoid overfitting, and why?)

Key Assessment Areas:

- Understanding of overfitting and its symptoms
- Use of appropriate techniques (e.g., regularization, feature reduction)
- Evidence of implementation and impact on model performance.

5. Results Analysis

a) Would you recommend your model for real-world deployment?

b) Is any model underfitting? If yes, what could be the possible reasons?

Key Assessment Areas:

- Interpretation of results and model behavior
- Use of model for prediction on unseen/test data
- Discussion of underfitting or overfitting issues
- Clarity, structure, and professionalism in presentation

Grading Matrix

Criteria	Excellent (70–100%)	Good (60–69%)	Satisfactory (50–59%)	Needs Improvement (40–49%)	Fail (0–39%)
1. Data Preparation [20%]	Dataset is appropriate and well justified. Data cleaning, feature encoding, normalization, and exploration are thorough and clearly explained.	Good preparation; most essential steps are present but could be explained more clearly.	Basic preparation shown, some preprocessing steps may be missing or unclear.	Limited effort in preparing the data; explanation lacks depth.	No meaningful preparation; incorrect or reused dataset; major issues.
2. Hyperparameter Tuning [20%]	Key hyperparameters are well chosen and justified. An appropriate tuning strategy (e.g., GridSearchCV) is used and	Good parameter selection and tuning. Explanation mostly clear with some minor gaps.	Some tuning performed, but rationale is minimal or unclear.	Minimal or ineffective tuning, with vague or generic explanation.	No tuning done, or incorrect application.

	clearly explained.				
3. Evaluation Metric [20%]	Appropriate metric(s) chosen for classification (e.g., precision , Recall, F1). Clear justification for selection and strong understanding shown.	Metric is suitable, but justification could be more detailed. Understanding is solid but not deep.	A metric is used with limited justification or understanding.	Metric applied incorrectly or without explanation.	No metric used or inappropriate metric applied.
4. Overfitting Avoidance [20%]	Clear evidence of model validation strategy (e.g., cross-validation, regularization). Techniques are appropriate and well justified.	Reasonable technique applied with some explanation. Minor gaps in clarity.	Overfitting considered but only at a surface level.	Weak understanding of overfitting; ineffective technique applied.	No consideration or strategy for overfitting.
5. Results Analysis & Deployment Recommendation [20%]	Strong analysis of both models. Realistic deployment recommendation and thoughtful discussion of underfitting.	Good insights with reasonable recommendation. Minor omissions or limited depth.	Results discussed but without strong conclusions. Underfitting discussed briefly.	Basic or superficial analysis with unclear conclusions.	No meaningful analysis or understanding of model performance.

GenAI usage

1	2	3	4	5
NO AI	AI-ASSISTED IDEA GENERATION AND STRUCTURING	AI-ASSISTED EDITING	AI TASK COMPLETION, HUMAN EVALUATION	FULL AI
The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their knowledge, understanding, and skills. AI must not be used at any point during the assessment.	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. No AI content is allowed in the final submission.	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. AI can be used, but your original work with no AI content must be provided in an appendix.	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. You may use AI throughout your assessment to support your own work and do not have to specify which content is AI generated

Performance Descriptor Table

Performance Descriptor Table		Evaluative Criteria/Performance Level Descriptors (what does performance at this level look like?)				
Scale	Grade Band	Criterion 1 – Data Preparation	Criterion 2 – Hyperparameter Tuning	Criterion 3 – Evaluation Metric	Criterion 4 – Overfitting Avoidance	Criterion 5 – Results Analysis
Exceptional	90–100%	Dataset is excellent and original; comprehensive preprocessing, justification, and feature engineering.	Advanced tuning approach with full justification and optimal model performance.	Metric perfectly suited, clearly justified, and well-implemented.	Sophisticated strategy (e.g., cross-validation with regularization); clearly justified.	Excellent analysis with strong real-world recommendation and insightful underfitting diagnosis.
Outstanding	80–89%	Very strong preparation and well-argued dataset choice; all preprocessing clearly described.	Well-chosen parameters and tuning strategy; strong understanding shown.	Metric is appropriate and reasoning is well explained.	Strong avoidance method; well implemented and justified.	Clear recommendation and thoughtful analysis of model behaviour.
Excellent	70–79%	Good dataset and preparation; most essential steps addressed well.	Appropriate tuning and explanation, but may lack depth.	Metric is suitable and applied correctly with explanation.	Good technique used; rationale could be more complete.	Sound analysis and logical recommendation made.
Very Good	60–69%	Preparation is mostly complete; minor gaps in reasoning or application.	Tuning done reasonably; some explanation given.	Metric is applied but with limited justification.	Overfitting addressed with basic techniques.	Analysis shows effort, but lacks detail in recommendation or diagnosis.
Good	50–59%	Basic data prep with standard steps; lacks clear explanation or dataset rationale.	Limited tuning; basic use of defaults with minor tuning.	Metric mentioned but unclear why it was chosen.	Overfitting touched on but not convincingly handled.	Minimal reflection on results or vague recommendation .
Sufficient	40–49%	Weak justification of dataset; poor or unclear preprocessing.	Minimal effort to tune or explain parameters.	Metric applied incorrectly or without clear understanding.	Ineffective or generic overfitting response.	Surface-level result discussion; recommendation missing or weak.

Somewhat / Unfulfilled	30–39%	Inadequate dataset; poor preparation or understanding.	Little to no parameter tuning.	Misunderstanding of evaluation metrics.	Confusion around overfitting; wrong or missing technique.	Incomplete or confused analysis; no conclusion.
Unfulfilled	0–29%	No dataset justification; no prep shown.	No evidence of tuning.	No metrics used or entirely incorrect.	No strategy applied.	No analysis or recommendation.
MIMLO Alignment (see Module Guide for Minimum Intended Learning Outcomes for Module)						
MIMLO to which criterion is aligned:		LO1	LO1, LO2	LO2	LO3	LO2, LO3

General Requirements for Students:

1. All assignments must be submitted no later than the stated deadline (date and time).
2. Assignments submitted after the latest deadline specified (including any approved extension deadline) are considered late and penalised according to the Quality Assurance Handbook (QAH) Part B Section 5.2.2.6 as follows:
 - a. A penalty of 2 marks will be applied per day or part thereof (including weekends and public holidays) for an ongoing failure to submit beyond the submission deadline.
 - b. An examiner has the right to refuse to mark the assignment if the submission instructions have not been observed.
 - c. Where a late assessment is submitted within 14 days of the deadline, and is of a passing standard, the late penalty is capped (such that the minimum grade that can be awarded is 40% for the late submission).
 - d. Where a late assessment is submitted more than 14 days after the deadline, it will receive 0%. The lecturer may, at their discretion, review the submission for feedback.
 - e. Where the assessment is undertaken in a group, the piece of work should be submitted in its complete entirety, and any penalty for late submission incurred applies to all group members.
3. Extensions to assignment submission deadlines will not be granted, other than in exceptional circumstances. To apply for an extension please go to <https://students.dbs.ie/dashboard/SCCM> and open a ticket.
4. All relevant provisions of the Assessment Regulations must be complied with (see QAH B.5).
 - a. Students are required to refer to the assessment regulations in their Programme Handbook, and on the Student Website.
 - b. Dublin Business School penalises students who engage in academic impropriety (i.e. plagiarism, collusion and/or copying, ghost writing/ essay mills, improper use of Generative Artificial Intelligence software).
 - i. Refer to the College's Generative AI Guidelines HERE for further information.

- c. Guides on referencing are available on the Library website: <https://libguides.dbs.ie/referencing>
 - d. Text-matching analysis software is integrated in Moodle to generate a report regarding the degree of text-matching in a submission.
5. Students are required to retain a copy of each assignment submitted, until the issuing of a transcript indicating the mark awarded and the closure of the Appeal period (2 weeks following the release of final results).
- a. Results can only be appealed following the release of final results, and the Appeal form must be submitted to the Exams Office within the Appeal period.
 - b. An appeal must be based on valid grounds (see the Appeals Policy QAH B.3.5), dissatisfaction with a grade is not sufficient grounds for an appeal.
 - c. Assignments must be appropriately packaged and presented.
 - d. All assignments should be submitted to your subject/course page on Moodle by the deadline date.
 - e. Where a submission involves digital media (i.e formats other than Word, Powerpoint or PDF), it is the submitting students' responsibility to ensure the media is appropriately labelled, fully working and they must retain a copy.
 - f. Components of an assessment which are not included in the final submission cannot normally be subsequently accepted for grading. It is the student's responsibility to ensure their file is uploaded correctly.
 - g. Include an electronic **cover sheet** with the following details to the front of the assignment (see below)
6. Assignments that *breach* the word count requirements will be penalised. *There is a 10% discretion, either way, applicable in terms of word count.*
7. When you submit your assignment you will be asked to click on a button which will declare the following:

By ticking this box I am confirming that this assignment/exam is all my own work. Any sources used have been referenced.

I have read the College rules regarding plagiarism in the QAH Part B Section 3 and understand that penalties will be applied accordingly if work is found not to be my own. All work uploaded is submitted via Ouriginal, whereby a text-matching report will show any similarities with other texts.